Kernix 0.8.0 / Performance brief

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

21 workloads

88 observations

Source precision retained

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f64 · 1 worker · allocated output · self product



How to read this report

Allocation-inclusive API latency unless marked reused output. Peer medians are observations.

Charts round to 6 significant digits (≈). The exact-value appendix preserves all recorded digits.

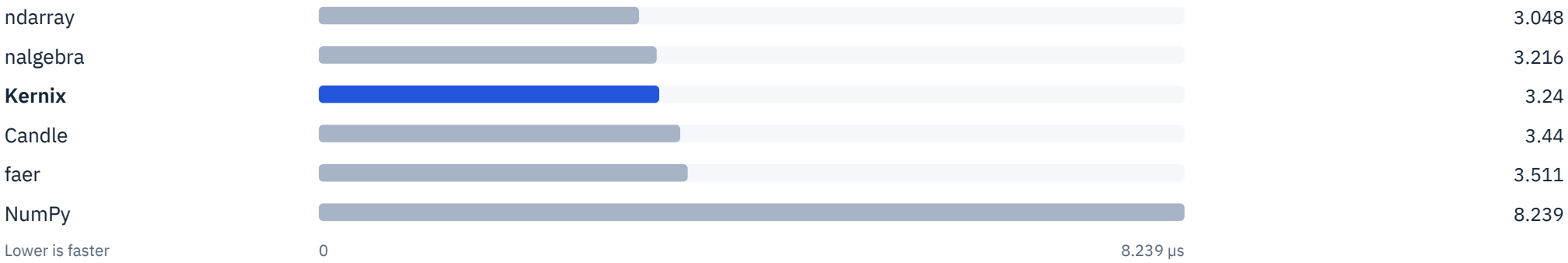
Kernix 0.8.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

32 × 32 · 32 × 32 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f64 · 1 worker · allocated output · self product



Kernix 0.8.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

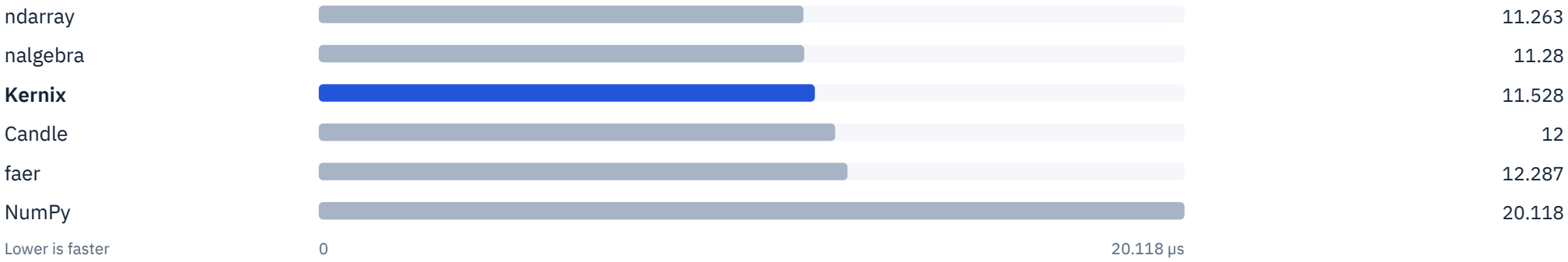
256 × 256 · 256 × 256 · f64 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

64 × 64 · 64 × 64 · f32 · 1 worker · allocated output · self product



Kernix 0.8.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Matrix multiplication

Per-call median

128 × 128 · 128 × 128 · f32 · 1 worker · allocated output · self product



Matrix multiplication

Per-call median

256 × 256 · 256 × 256 · f32 · 1 worker · allocated output · self product

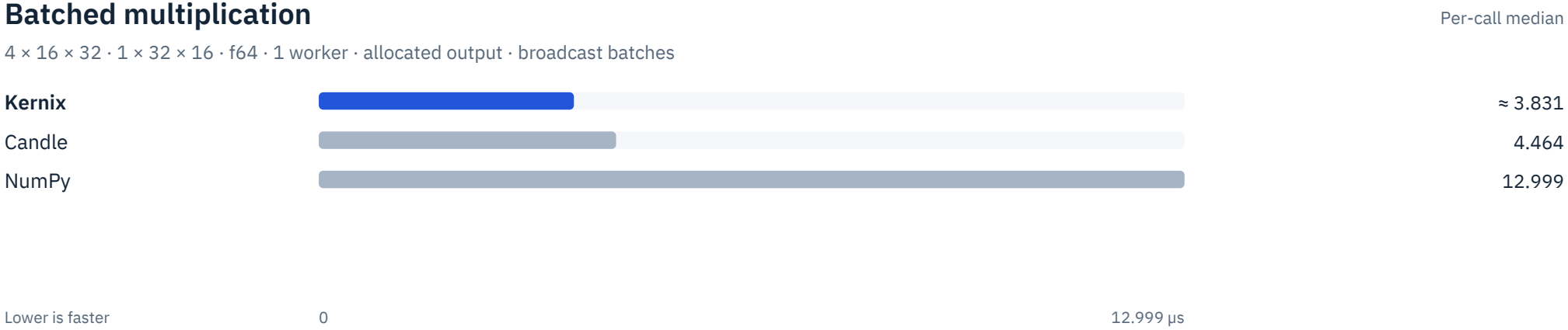


Kernix 0.8.0 / Matrix multiplication

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Batched multiplication

4 × 16 × 32 · 1 × 32 × 16 · f64 · 1 worker · allocated output · broadcast batches



Kernix 0.8.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

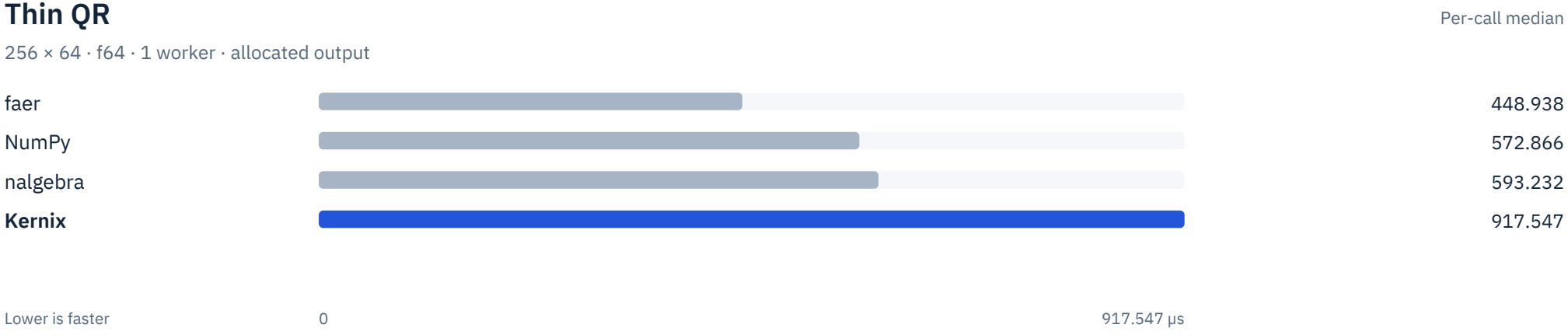
Thin QR

96 × 32 · f64 · 1 worker · allocated output



Thin QR

256 × 64 · f64 · 1 worker · allocated output



Kernix 0.8.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

64 × 64 · f64 · 1 worker · allocated output



Cholesky factorization

128 × 128 · f64 · 1 worker · allocated output



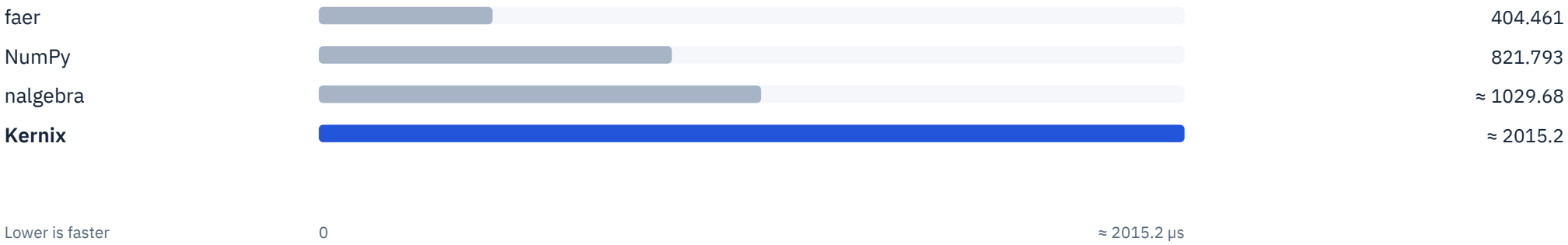
Kernix 0.8.0 / Decompositions & solvers

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Cholesky factorization

256 × 256 · f64 · 1 worker · allocated output

Per-call median



Kernix 0.8.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Record + backward

Per-call median

sum(A · W), 32 × 16 · 16 × 8 · f64 · 1 worker · allocated output · self product



Model inference

Per-call median

batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



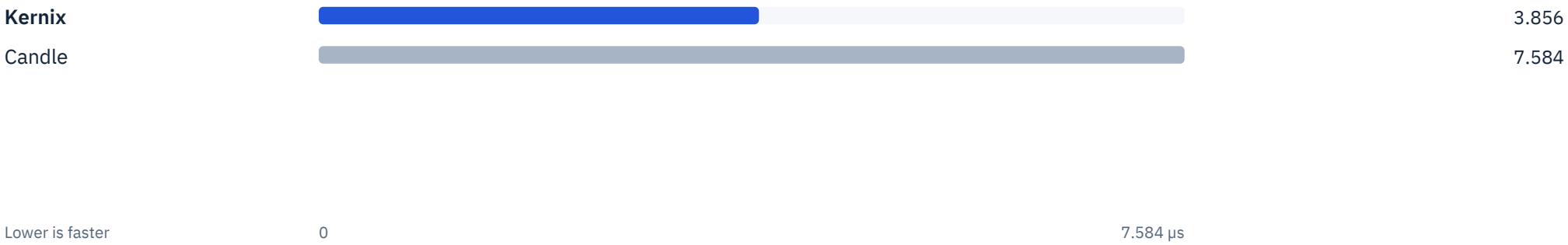
Kernix 0.8.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Model recording + forward

Per-call median

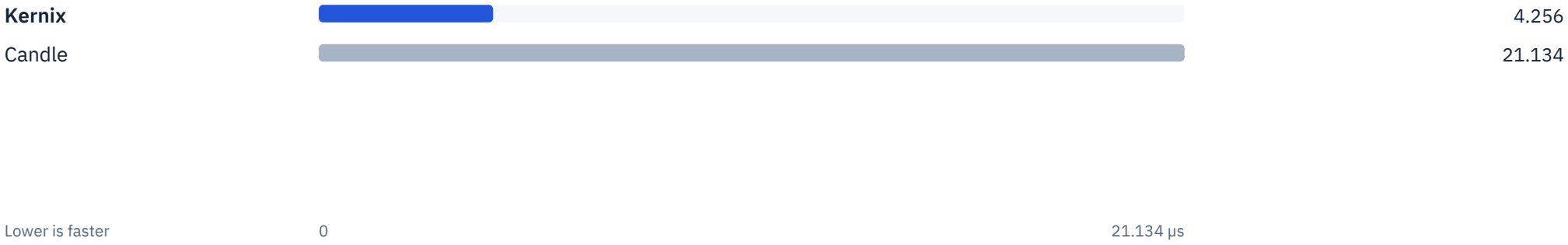
batched normalized/sliced MLP · f64 · 1 worker · preparation as documented



Model backward

Per-call median

prepared graph → fresh gradients · f64 · 1 worker · prepared graph · fresh gradients



Kernix 0.8.0 / Training & autodiff

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Parameter update

transactional SGD, fixed gradients · f64 · 1 worker · preparation as documented



Kernix 0.8.0 / Statistics & activations

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Row softmax

256 × 64 · f64 · 1 worker · allocated output



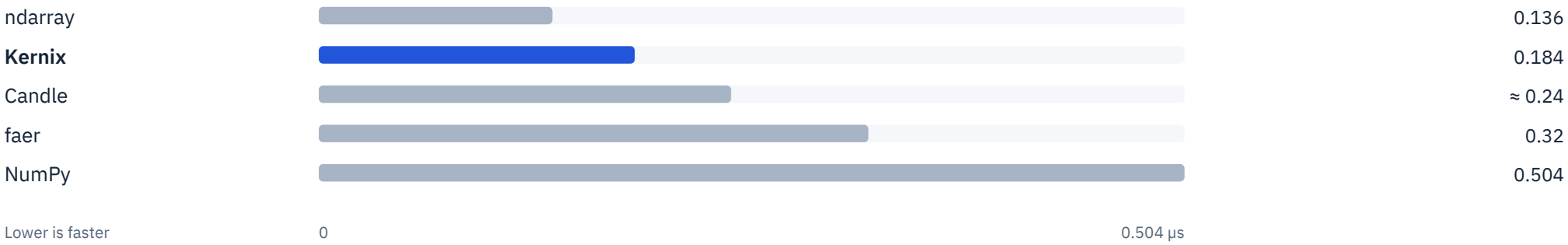
Kernix 0.8.0 / Tensor construction

Neoverse-N2 · Per-call median · µs · NumPy BLAS: scipy-openblas

Create zeros

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Create ones

Per-call median

64 × 64 · f32 · 1 worker · allocated output



Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / Kernix 32 × 32 · 32 × 32 · f64	3240	3.24
Matrix multiplication / ndarray 32 × 32 · 32 × 32 · f64	3048	3.048
Matrix multiplication / faer 32 × 32 · 32 × 32 · f64	3511	3.511
Matrix multiplication / Candle 32 × 32 · 32 × 32 · f64	3440	3.44
Matrix multiplication / NumPy 32 × 32 · 32 × 32 · f64	8239	8.239
Matrix multiplication / nalgebra 32 × 32 · 32 × 32 · f64	3216	3.216
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f64	22486	22.486
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f64	31022	31.022
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f64	22174	22.174
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f64	21799	21.799

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / faer 64 × 64 · 64 × 64 · f64	22630	22.63
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f64	22559	22.559
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f64	225025	225.025
Matrix multiplication / faer 128 × 128 · 128 × 128 · f64	169485	169.485
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f64	224513	224.513
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f64	225569	225.569
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f64	194635	194.635
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f64	168765	168.765
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f64	1329142	1329.142
Matrix multiplication / faer 256 × 256 · 256 × 256 · f64	1621235	1621.235

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f64	1463590	1463.59
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f64	1332135	1332.135
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f64	1321904	1321.904
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f64	1324464	1324.464
Matrix multiplication / nalgebra 64 × 64 · 64 × 64 · f32	11280	11.28
Matrix multiplication / Candle 64 × 64 · 64 × 64 · f32	12000	12
Matrix multiplication / Kernix 64 × 64 · 64 × 64 · f32	11528	11.528
Matrix multiplication / NumPy 64 × 64 · 64 × 64 · f32	20118	20.118
Matrix multiplication / faer 64 × 64 · 64 × 64 · f32	12287	12.287
Matrix multiplication / ndarray 64 × 64 · 64 × 64 · f32	11263	11.263

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / ndarray 128 × 128 · 128 × 128 · f32	97337	97.337
Matrix multiplication / NumPy 128 × 128 · 128 × 128 · f32	99833	99.833
Matrix multiplication / nalgebra 128 × 128 · 128 × 128 · f32	97426	97.426
Matrix multiplication / Kernix 128 × 128 · 128 × 128 · f32	97833	97.833
Matrix multiplication / Candle 128 × 128 · 128 × 128 · f32	85922	85.922
Matrix multiplication / faer 128 × 128 · 128 × 128 · f32	87203	87.203
Matrix multiplication / faer 256 × 256 · 256 × 256 · f32	677707	677.707
Matrix multiplication / Candle 256 × 256 · 256 × 256 · f32	724783	724.783
Matrix multiplication / nalgebra 256 × 256 · 256 × 256 · f32	795611	795.611
Matrix multiplication / Kernix 256 × 256 · 256 × 256 · f32	802778	802.778

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Matrix multiplication / NumPy 256 × 256 · 256 × 256 · f32	727023	727.023
Matrix multiplication / ndarray 256 × 256 · 256 × 256 · f32	799795	799.795
Batched multiplication / Candle 4 × 16 × 32 · 1 × 32 × 16 · f64	4464	4.464
Batched multiplication / NumPy 4 × 16 × 32 · 1 × 32 × 16 · f64	12999	12.999
Batched multiplication / Kernix 4 × 16 × 32 · 1 × 32 × 16 · f64	3831.0000000000005	3.8310000000000005
Thin QR / Kernix 96 × 32 · f64	89546	89.546
Thin QR / faer 96 × 32 · f64	65395.00000000001	65.39500000000001
Thin QR / nalgebra 96 × 32 · f64	60876	60.876
Thin QR / NumPy 96 × 32 · f64	65963	65.963
Thin QR / Kernix 256 × 64 · f64	917547	917.547

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Thin QR / faer 256 × 64 · f64	448938	448.938
Thin QR / nalgebra 256 × 64 · f64	593232	593.232
Thin QR / NumPy 256 × 64 · f64	572866	572.866
Cholesky factorization / nalgebra 64 × 64 · f64	20359	20.359
Cholesky factorization / Kernix 64 × 64 · f64	27326	27.326
Cholesky factorization / faer 64 × 64 · f64	13231	13.231
Cholesky factorization / NumPy 64 × 64 · f64	18143	18.143
Cholesky factorization / NumPy 128 × 128 · f64	85178	85.178
Cholesky factorization / nalgebra 128 × 128 · f64	128694.99999999999	128.69499999999999
Cholesky factorization / faer 128 × 128 · f64	65011.99999999999	65.01199999999999

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Cholesky factorization / Kernix 128 × 128 · f64	279557	279.557
Cholesky factorization / faer 256 × 256 · f64	404461	404.461
Cholesky factorization / nalgebra 256 × 256 · f64	1029683	1029.683
Cholesky factorization / Kernix 256 × 256 · f64	2015201	2015.201
Cholesky factorization / NumPy 256 × 256 · f64	821793	821.793
Record + backward / Kernix sum(A · W), 32 × 16 · 16 × 8 · f64	3527	3.527
Record + backward / Candle sum(A · W), 32 × 16 · 16 × 8 · f64	7223	7.223
Model inference / Candle batched normalized/sliced MLP · f64	5320	5.32
Model inference / Kernix batched normalized/sliced MLP · f64	3023	3.023
Model recording + forward / Kernix batched normalized/sliced MLP · f64	3856	3.856

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Model recording + forward / Candle batched normalized/sliced MLP · f64	7584	7.584
Model backward / Candle prepared graph → fresh gradients · f64	21134	21.134
Model backward / Kernix prepared graph → fresh gradients · f64	4256	4.256
Parameter update / Kernix transactional SGD, fixed gradients · f64	728	0.728
Row softmax / NumPy 256 × 64 · f64	135151	135.151
Row softmax / Candle 256 × 64 · f64	110969	110.969
Row softmax / Kernix 256 × 64 · f64	113168	113.168
Row softmax / ndarray 256 × 64 · f64	97410	97.41
Create zeros / Kernix 64 × 64 · f32	184	0.184
Create zeros / faer 64 × 64 · f32	320	0.32

Recorded precision / exact latency table

Neoverse-N2 · Per-call median · no rounding applied

WORKLOAD / LIBRARY	NANOSECONDS	MICROSECONDS
Create zeros / ndarray 64 × 64 · f32	136	0.136
Create zeros / NumPy 64 × 64 · f32	504	0.504
Create zeros / Candle 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / faer 64 × 64 · f32	239.99999999999997	0.23999999999999997
Create ones / Kernix 64 × 64 · f32	256	0.256
Create ones / Candle 64 × 64 · f32	344	0.344
Create ones / NumPy 64 × 64 · f32	1528	1.528
Create ones / ndarray 64 × 64 · f32	208	0.208